Supplement to "Identifying Rational Types in Unknown Environments"

Appendix A: Microstructure and Competing Hazards in Colombia

August 17, 2026

This supplement presents in full detail the microeconometric results summarized in Section 4.1 of the main paper ("Empirical anchoring of the parameter space"). It documents the CNMH/SIEVCAC data, the construction of the analytical sample, the sample-selection test, the multinomial logit (MNL) model of kidnapping outcomes, specification diagnostics and robustness checks, and the cause-specific Cox proportional hazards models of resolution timing. All estimates were reproduced from the Stata scripts in the `appendix_A_micro/scripts/` replication package (Section S7).

S1 Data and sample construction

Individual kidnapping records come from the Information System on Events of Violence in the Colombian Armed Conflict (SIEVCAC), maintained by the National Center of Historical Memory (CNMH) (CNMH, 2026, 2013). The replication package includes the merged case–victim file `stata/data/Data_merge.dta` (N = 39,369 victim-level records) and the

Table S1.1: Analytical Sample Construction

Stage	n	Share of prior stage
Universe (merged CNMH records)	39,369	—
Valid event date	36,916	93.8%
Extortionate typology	11,270	30.5%
Observable outcome	1,133	10.1%
Complete covariates	1,125	99.3%

Note. Valid event date: non-missing `Event_Date` and `Year`$\neq 0$ after parsing `Data_merge.dta`. Extortionate share of universe: 11,270/39,369 = 28.6%. Estimation share: 1,125/39,369 = 2.9%. Eight cases with missing geographic zone are dropped at the final stage. *Source:* Authors' own calculations based on CNMH/SIEVCAC microdata (Section S1). Software: Stata 19 (StataCorp, 2025).

estimation-ready checkpoint `stata/data/analytical_sample.dta` (n = 1,125 after outcome filters, variable construction, and dropping eight cases with missing geographic zone).

S1.1.0.0 Universe and sequential filters

Table S1.1 summarizes the sample flow reported in the main text. The universe comprises all documented kidnapping victims in the merged CNMH export. The first filter retains extortionate kidnappings (`Kidnap_Type` = ``EXTORSIVO'' in the source data) with a valid event date. The second filter requires an observable liberation outcome (excluding not-observed, other, and non-applicable codes in `Release_Type`). The third filter requires complete covariates for estimation; eight cases with missing geographic zone are dropped, yielding n = 1,125.

S1.2.0.0 Outcome variable

The dependent variable `Y_Outcome` classifies each extortionate episode into four mutually exclusive outcomes (reference category: Payment):

- **Payment** ($Y = 1$): ransom paid, hostage exchange, or intermediated payment.
- **Rescue** ($Y = 3$): liberation by state security forces.

Table S1.2: Outcome Distribution in the Estimation Sample

Outcome	Count	Share
Payment (reference)	512	45.5%
Rescue	471	41.9%
Escape or release without payment	121	10.8%
Death in captivity	21	1.9%
Total	1,125	100.0%

Note. Estimation sample $n = 1{,}125$. Payment is the MNL reference category. *Source:* Authors' own calculations based on CNMH/SIEVCAC microdata (Section S1). Software: Stata 19 (StataCorp, 2025).

- **Escape or release without payment** ($Y = 0$): flight or release under external pressure without ransom.

- **Death** ($Y = 2$): victim killed or found dead in captivity.

Table S1.2 reports the distribution in the estimation sample.

S1.3.0.0 Covariates

The covariate vector $\mathbf{X}_{i,m} \in \mathbb{R}^{26}$ in the main text comprises four blocks (all coded as indicator variables with stated reference categories):

1. **Perpetrator** ($\mathbf{G}_i$): FARC (ref.), ELN, paramilitaries, common delinquency/other.

2. **Victim profile** ($\mathbf{V}_i$): high income (ref.), middle class, vulnerable, public sector, unknown.

3. **Context** ($\mathbf{C}_{i,m}$): geographic zone (metropolis ref., Andean, Caribbean, Pacific/conflict, East/jungle) and historical period (pre-1998 ref., 1998–2002, 2003–2010, post-2011).

4. **Tactics** ($\mathbf{T}_i$): modus operandi, collective vs. individual structure, GAULA intervention, victim sex, captivity duration category (express ≤ 2 days ref., short ≤ 30 days, long > 30 days, missing), and ln(total victims per case).

Table S2.1: Attrition Logistic Regression Results

Covariate	OR	p-value	Significance
Non-metropolitan zones	1.9–2.2	< 0.001	1%
Period 1998–2002	1.76	0.001	1%
Victim profile	—	> 0.10	—
ln(total victims per case)	—	> 0.10	—

Note. Estimated on extortionate cases with complete attrition covariates ($n = 11{,}253$; 11,270 extortionate cases with valid date). Stata drops 17 observations with missing covariates. Cluster-robust standard errors by municipality. OR > 1 indicates higher odds of an unobservable outcome. Non-metropolitan zones comprise Andean, Caribbean, Pacific, and East/jungle categories. Victim-profile and scale covariates are not predictive at the 10% level; geography and the 1998–2002 period show systematic attrition. The Significance column reports the highest conventional level (1%, 5%, 10%); "—" denotes not significant at 10%. *Source:* Authors' own calculations based on CNMH/SIEVCAC microdata (Section S1). Software: Stata 19 (StataCorp, 2025).

Standard errors are clustered at the municipal level (373 clusters). Variable construction rules are implemented in `microstructure_replication.do` and `sample_selection_robustness.do`.

S2 Sample-selection test

Because only 10.1% of extortionate cases have observable outcomes, a binary logistic model was estimated on the extortionate stratum with complete attrition covariates ($n = 11{,}253$ of 11,270 extortionate cases with valid date), with dependent variable `Missing_Outcome`=1 when the liberation type is unobserved (Little and Rubin, 2020). Table S2.1 summarizes the main findings as odds ratios with cluster-robust standard errors by municipality.

The test rejects complete randomness of missing outcomes while supporting the internal validity of the structural estimators: substantive covariates do not predict missingness, and the remaining bias is territorial (metropolitan overrepresentation).

S3 Multinomial logit model

S3.1.0.0 Specification

With Payment as the base category ($Y = 1$), the MNL estimates

$$\ln\left(\frac{\Pr(Y_{i,m} = j \mid \mathbf{X}_{i,m})}{\Pr(Y_{i,m} = 1 \mid \mathbf{X}_{i,m})}\right) = \alpha_j + \mathbf{X}'_{i,m}\boldsymbol{\beta}_j, \quad j \in \{0, 2, 3\}. \tag{S3.1}$$

The main specification uses cluster-robust standard errors at the municipality level (Wooldridge, 2010).

S3.2.0.0 Model fit

The model is jointly significant: Wald $\chi^2(78) = 5{,}250.34$, $p < 0.001$; McFadden pseudo-$R^2 = 0.18$.

S3.3.0.0 Full estimation results

Table S3.1 reports the complete MNL specification: all 26 covariates for the three non-payment outcomes (Escape, Death, Rescue), with cluster-robust standard errors, z-statistics, p-values, and 95% confidence intervals. The baseline outcome is Payment ($Y = 1$).

Table S3.1: Multinomial Logistic Regression Results (Full Model)

Outcome / Predictors	**RRR**	**Std. Err.**	z	$P > \|z\|$	**[95% Conf.**	**Interval]**
Outcome: Escape or release without payment						
Perpetrator group (ref. FARC)						
ELN	0.800	0.284	−0.63	0.531	0.399	1.606
Paramilitaries	0.932	0.419	−0.16	0.876	0.386	2.252

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Table S3.1 — continued from previous page

Outcome / Predictors	RRR	Std. Err.	z	$P > \|z\|$	[95% Conf. Interval]	
Common delinquency/other	0.493	0.233	−1.50	0.135	0.195	1.246
Victim profile (ref. upper class)						
Middle class	1.868	0.749	1.56	0.119	0.852	4.097
Vulnerable population	0.624	0.278	−1.06	0.291	0.260	1.496
Public sector	3.917	1.842	2.90	0.004	1.559	9.844
Unknown	1.469	0.519	1.09	0.276	0.736	2.935
Geographic zone (ref. metropolis)						
Andean	0.637	0.284	−1.01	0.313	0.266	1.528
Caribbean	0.496	0.238	−1.46	0.143	0.194	1.268
Pacific/conflict zone	0.554	0.269	−1.22	0.224	0.214	1.436
East/jungle	0.727	0.399	−0.58	0.561	0.248	2.133
Modus operandi (ref. direct attack)						
Illegal roadblock	2.287	1.187	1.59	0.111	0.827	6.323
Military incursion	1.274	0.627	0.49	0.622	0.486	3.341
Deception	0.878	0.483	−0.24	0.814	0.299	2.583
Unknown	1.444	0.438	1.21	0.226	0.797	2.615
Kidnapping structure						
Group structure (vs. individual)	0.760	0.308	−0.68	0.498	0.343	1.683
GAULA intervention						
GAULA intervention (vs. none)	0.942	0.350	−0.16	0.873	0.455	1.952
Victim sex (ref. male)						
Female	0.625	0.260	−1.13	0.259	0.276	1.414
No information	2.038	3.405	0.43	0.670	0.077	53.832

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Outcome / Predictors	RRR	Std. Err.	z	$P > \|z\|$	[95% Conf. Interval]	
Duration (ref. express $\leq$ 2 days)						
Short ($\leq$ 30 days)	0.322	0.198	−1.85	0.065	0.096	1.072
Long ($>$ 30 days)	0.235	0.156	−2.18	0.029	0.064	0.861
Missing duration	0.464	0.282	−1.26	0.207	0.141	1.529
ln(total victims per case)	0.764	0.197	−1.04	0.297	0.461	1.267
Historical period (ref. pre-1998)						
Crisis/Caguán (1998–2002)	1.506	0.675	0.91	0.360	0.626	3.624
Democratic security (2003–2010)	1.235	0.587	0.44	0.658	0.486	3.135
Peace process/current (2011+)	11.283	7.227	3.78	0.000	3.215	39.596
Constant	0.486	0.401	−0.87	0.382	0.097	2.447
Outcome: Death in captivity						
Perpetrator group (ref. FARC)						
ELN	4.500	6.741	1.00	0.315	0.239	84.787
Paramilitaries	20.306	25.564	2.39	0.017	1.722	239.473
Common delinquency/other	63.629	85.557	3.09	0.002	4.562	887.547
Victim profile (ref. upper class)						
Middle class	0.515	0.876	−0.39	0.696	0.018	14.455
Vulnerable population	7.056	5.195	2.65	0.008	1.667	29.873
Public sector	0.000	0.000	−13.09	0.000	0.000	0.000
Unknown	1.219	0.998	0.24	0.809	0.245	6.062
Geographic zone (ref. metropolis)						
Andean	0.560	0.604	−0.54	0.591	0.067	4.643
Caribbean	0.000	0.000	−15.61	0.000	0.000	0.000

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Table S3.1 — continued from previous page

Outcome / Predictors	RRR	Std. Err.	z	$P > \|z\|$	[95% Conf.	Interval]
Pacific/conflict zone	0.456	0.521	−0.69	0.492	0.049	4.284
East/jungle	0.624	0.733	−0.40	0.688	0.062	6.246
Modus operandi (ref. direct attack)						
Illegal roadblock	0.000	0.000	−16.21	0.000	0.000	0.000
Military incursion	0.000	0.000	−22.26	0.000	0.000	0.000
Deception	0.537	0.698	−0.48	0.632	0.042	6.863
Unknown	1.244	0.736	0.37	0.712	0.390	3.967
Kidnapping structure						
Group structure (vs. individual)	11.744	8.468	3.42	0.001	2.858	48.262
GAULA intervention						
GAULA intervention (vs. none)	0.536	0.512	−0.65	0.514	0.082	3.489
Victim sex (ref. male)						
Female	2.450	1.831	1.20	0.230	0.566	10.600
No information	47.771	83.378	2.22	0.027	1.561	1461.557
Duration (ref. express ≤ 2 days)						
Short (≤ 30 days)	0.580	0.546	−0.58	0.562	0.092	3.666
Long (> 30 days)	0.477	0.545	−0.65	0.517	0.051	4.474
Missing duration	0.361	0.420	−0.88	0.381	0.037	3.522
ln(total victims per case)	0.084	0.054	−3.90	0.000	0.024	0.293
Historical period (ref. pre-1998)						
Crisis/Caguán (1998–2002)	1.230	1.215	0.21	0.834	0.177	8.529
Democratic security (2003–2010)	1.405	1.397	0.34	0.732	0.200	9.856
Peace process/current (2011+)	10.008	9.700	2.38	0.017	1.497	66.890

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Outcome / Predictors	RRR	Std. Err.	z	$P > \|z\|$	[95% Conf. Interval]	
Constant	0.004	0.010	−1.98	0.047	0.000	0.932
Outcome: Rescue						
Perpetrator group (ref. FARC)						
ELN	0.840	0.198	−0.74	0.459	0.529	1.334
Paramilitaries	0.680	0.250	−1.05	0.295	0.330	1.399
Common delinquency/other	0.494	0.174	−2.00	0.045	0.247	0.984
Victim profile (ref. upper class)						
Middle class	1.277	0.334	0.93	0.350	0.765	2.130
Vulnerable population	0.800	0.221	−0.81	0.420	0.465	1.376
Public sector	0.654	0.281	−0.99	0.323	0.282	1.518
Unknown	0.749	0.197	−1.10	0.272	0.447	1.255
Geographic zone (ref. metropolis)						
Andean	0.800	0.329	−0.54	0.587	0.357	1.792
Caribbean	0.426	0.194	−1.87	0.061	0.174	1.041
Pacific/conflict zone	0.656	0.311	−0.89	0.374	0.259	1.661
East/jungle	0.740	0.332	−0.67	0.501	0.307	1.782
Modus operandi (ref. direct attack)						
Illegal roadblock	1.513	0.667	0.94	0.348	0.638	3.589
Military incursion	1.112	0.388	0.31	0.760	0.561	2.205
Deception	1.232	0.559	0.46	0.645	0.507	2.998
Unknown	2.077	0.518	2.93	0.003	1.274	3.386
Kidnapping structure						
Group structure (vs. individual)	0.972	0.280	−0.10	0.921	0.552	1.710

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Table S3.1 — continued from previous page

Outcome / Predictors	RRR	Std. Err.	z	$P > \|z\|$	[95% Conf.	Interval]
GAULA intervention						
GAULA intervention (vs. none)	1.390	0.481	0.95	0.341	0.706	2.737
Victim sex (ref. male)						
Female	1.894	0.415	2.92	0.004	1.233	2.910
No information	7.430	10.051	1.48	0.138	0.524	105.297
Duration (ref. express ≤ 2 days)						
Short (≤ 30 days)	0.183	0.085	−3.65	0.000	0.074	0.455
Long (> 30 days)	0.079	0.043	−4.69	0.000	0.027	0.228
Missing duration	0.448	0.171	−2.11	0.035	0.213	0.945
ln(total victims per case)	1.116	0.197	0.62	0.534	0.789	1.578
Historical period (ref. pre-1998)						
Crisis/Caguán (1998–2002)	1.028	0.345	0.08	0.935	0.532	1.986
Democratic security (2003–2010)	1.554	0.595	1.15	0.249	0.734	3.290
Peace process/current (2011+)	3.849	1.906	2.72	0.006	1.459	10.157
Constant	2.314	1.429	1.36	0.174	0.690	7.765
N			1,125			
Wald $\chi^2(78)$			5,250.34			
Prob $> \chi^2$			0.0000			
Pseudo R^2			0.1847			

Note. Relative risk ratios (RRR) reported. Baseline outcome is Payment. Cluster-robust standard errors at the municipality level (373 clusters; $n = 1{,}125$). p-values in parentheses are two-sided. Significance at 5%: $|z| > 1.96$. *Source:* Authors' own calculations based on CNMH/SIEVCAC microdata (Section S1). Software: Stata 19 (StataCorp, 2025).

Table S3.2: Selected Multinomial Logit Relative Risk Ratios

Outcome	Covariate	RRR	p-value	Significance
Escape	Post-2011 period	11.28	< 0.01	1%
Escape	Public-sector victim	3.92	< 0.01	1%
Rescue	Short captivity (≤ 30 d)	0.18	< 0.001	1%
Rescue	Long captivity (> 30 d)	0.08	< 0.001	1%

Note. Base category: Payment. RRR = $\exp(\hat{\beta})$. Duration categories are relative to express captivities (≤ 2 days). Period contrast: post-2011 vs. pre-1998. The Significance column reports the highest conventional level (1%, 5%, 10%); "—" denotes not significant at 10%. *Source:* Authors' own calculations based on CNMH/SIEVCAC microdata (Section S1). Software: Stata 19 (StataCorp, 2025).

S3.4.0.0 Key relative risk ratios

Table S3.2 summarizes the three headline findings in Section 4.1 of the main paper, expressed as relative risk ratios (RRR = $\exp(\hat{\beta})$). The corresponding rows in Table S3.1 are Peace process/current (Escape), public-sector victim (Escape), and short/long duration (Rescue).

S3.5.0.0 Predicted probabilities (margins)

Table S3.3 reports average marginal predicted probabilities from the main MNL (unconditional margins).

S4 MNL diagnostics and robustness

S4.1.0.0 IIA test

The Hausman–McFadden (Hausman and McFadden, 1984) SUEST generalization (Weesie, 1999) compares the full four-outcome model with a restricted model that excludes Death. The test does not reject independence of irrelevant alternatives: $\chi^2(1) = 0.24$, $p = 0.625$.

Table S3.3: Average Predicted Probabilities from the Main MNL

Dimension	Category	Pr(·)
Rescue by geographic zone		
	Metropolis	0.465
	Andean	0.447
	Pacific	0.414
	East/jungle	0.424
	Caribbean	0.343
Death by perpetrator group		
	FARC	0.002
	ELN	0.007
	Paramilitaries	0.026
	Common delinquency	0.071

Note. Unconditional margins from `margins` after the main MNL ($n = 1{,}125$). Rescue probabilities decline away from metropolitan nodes; death probabilities rise as logistical retention capacity falls. *Source:* Authors' own calculations based on CNMH/SIEVCAC microdata (Section S1). Software: Stata 19 (StataCorp, 2025).

Table S4.1: MNL Robustness: Death-Equation Coefficient for Perpetrator Group

	$\hat{\beta}$ (Death equation)		
Covariate	Main	Robust SE	No duration
Common delinquency (vs. FARC)	4.15	4.15	4.15

Note. Municipal clusters in Main and No duration; heteroskedasticity- robust standard errors in Robust SE. The log-relative-risk for common delinquency is identical across all three specifications ($\hat{\beta} = 4.15$, $p < 0.01$), confirming that the perpetrator-group effect in the Death equation is not an artifact of the variance estimator or the inclusion of duration categories. *Source:* Authors' own calculations based on CNMH/SIEVCAC microdata (Section S1). Software: Stata 19 (StataCorp, 2025).

S4.2.0.0 Coefficient stability

Three specifications are compared: (i) main model with municipal clusters; (ii) robust standard errors without clustering; and (iii) specification without duration categories.

Table S4.1 reports the key coefficient from the Death equation.

S4.3.0.0 Death-rate heterogeneity

Raw death rates in the estimation sample are 8.0% for common delinquency and 0.2% for the FARC. A two-sample proportion test (FARC vs. common delinquency) rejects equal rates: $z = -5.71$, $p < 0.001$.

S5 Cause-specific Cox proportional hazards

S5.1.0.0 Specification

Following Cox (1972) and the competing-risks framework of Bradburn et al. (2003), four cause-specific Cox models are estimated on `Days_Captivity` (days in captivity). In each model, one outcome defines the event of interest and the remaining three outcomes are treated as right-censored. The cause-specific hazard is

$$h_j(t \mid \mathbf{X}) = h_{0j}(t) \exp\left(\mathbf{X}'\tilde{\boldsymbol{\beta}}_j\right), \qquad j \in \{\text{Payment, Rescue, Death, Escape}\}, \tag{S5.1}$$

with the same covariate blocks as the MNL but excluding duration categories, which would be collinear with the time axis. The Cox subsample comprises $N = 461$ cases with positive captivity duration and complete covariates (602 cases with missing duration and 64 zero-day captivities are excluded; Breslow ties). FARC is the reference perpetrator group. Cluster-robust standard errors are reported.

S5.2.0.0 Estimation results

Table S5.1 reports all four cause-specific equations (Payment, Rescue, Death, and Escape/release without payment). Table S5.2 lists the coefficients emphasized in Section 4.1 of the main paper.

Table S5.1: Cause-Specific Cox Proportional Hazards Results (Four Equations)

Predictors	Payment (HR)	Rescue (HR)	Death (HR)	Escape (HR)
Perpetrator group (ref. FARC)				
ELN	1.186	1.057	3.039	0.569
	(0.317)	(0.837)	(0.404)	(0.225)
Paramilitaries	4.284***	1.262	12.211*	7.547***
	(0.000)	(0.530)	(0.049)	(0.000)
Common delinquency	1.713*	0.935	49.988**	1.157
	(0.021)	(0.862)	(0.004)	(0.812)
Victim profile (ref. upper class)				
Middle class	0.780	1.071	0.000***	0.680
	(0.274)	(0.823)	(0.000)	(0.467)
Vulnerable population	1.542*	0.512	13.108*	0.676
	(0.046)	(0.053)	(0.030)	(0.512)
Public sector	0.199***	0.237***	0.000***	2.133
	(0.000)	(0.001)	(0.000)	(0.218)
Geographic zone (ref. metropolis)				
Andean	1.290	1.709	0.126	0.445
	(0.450)	(0.465)	(0.052)	(0.209)
Caribbean	1.170	1.024	0.000***	0.523
	(0.649)	(0.975)	(0.000)	(0.304)
East/jungle	0.815	0.625	0.231	0.674
	(0.644)	(0.530)	(0.117)	(0.579)
Modus operandi (ref. direct attack)				
Illegal roadblock	1.780*	2.339	0.000***	5.044**

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Table S5.1 — continued from previous page

Predictors	Payment (HR)	Rescue (HR)	Death (HR)	Escape (HR)
	(0.012)	(0.069)	(0.000)	(0.003)
Deception	1.330	3.275***	0.000***	0.555
	(0.309)	(0.000)	(0.000)	(0.421)
Kidnapping characteristics				
Group structure (vs. individual)	0.950	0.577	3.736	3.226**
	(0.801)	(0.114)	(0.331)	(0.008)
GAULA intervention (vs. none)	0.716	0.597	1.172	0.976
	(0.267)	(0.244)	(0.932)	(0.961)
ln(total victims per case)	1.205	1.593*	0.659	0.295***
	(0.191)	(0.052)	(0.739)	(0.000)
Historical period (ref. pre-1998)				
Crisis/Caguán (1998–2002)	1.112	0.677	1.366	0.401
	(0.514)	(0.243)	(0.878)	(0.150)
Democratic security (2003–2010)	0.735	2.022*	9.186	0.220
	(0.125)	(0.036)	(0.108)	(0.063)
Peace process/current (2011+)	0.510*	4.863***	35.559**	4.886**
	(0.035)	(0.000)	(0.008)	(0.002)
Model statistics				
Observations (N)	461	461	461	461
Events	277	120	13	51
Wald χ^2	117.57	182.68	—	116.45
Prob > χ^2	0.0000	0.0000	0.0000	0.0000

Note. Hazard ratios (HR) with cluster-robust p-values in parentheses (two-sided). * $p < .05$; ** $p < .01$; *** $p < .001$. Duration categories are excluded (collinear with the time axis). The Cox subsample ($N = 461$) drops 602 cases with missing duration and 64 zero-day

Table S5.2: Selected Cause-Specific Cox Hazard Ratios

Equation	Covariate	HR	p-value	Significance
Payment equation				
	Paramilitaries	4.28	< 0.001	1%
	Common delinquency	1.71	0.021	5%
	ELN	1.19	0.317	—
Payment and Rescue equations				
	Public-sector victim (Payment)	0.20	< 0.001	1%
	Public-sector victim (Rescue)	0.24	0.001	1%
Post-2011 period				
	Post-2011 (Payment)	0.51	0.035	5%
	Post-2011 (Rescue)	4.86	< 0.001	1%

Note. Rows match Table S5.1 (rounded for the narrative). Estimated on the Cox subsample ($N = 461$) which drops 602 cases with missing duration and 64 zero-day captivities. Reference perpetrator group: FARC. HR $= \exp(\tilde{\beta})$; HR > 1 indicates faster resolution, HR < 1 slower resolution. The Significance column reports the highest conventional level (1%, 5%, 10%); "—" denotes not significant at 10%. Payment speed falls 49% post-2011 ($1 - 0.51$); Rescue hazard multiplies by $\approx$4.86. *Source:* Authors' own calculations based on CNMH/SIEVCAC microdata (Section S1). Software: Stata 19 (StataCorp, 2025).

captivities (Breslow ties). Payment and Rescue coefficients match Section 2.3 of the main paper; Death and Escape equations use the same covariate vector (`cox_only.do`). The Death equation has only 13 events: several HRs hit numerical bounds (reported as 0.000) owing to quasi-complete separation; its Wald statistic is omitted from interpretation. Pacific zone, unknown victim profile, military incursion, unknown modus, and sex indicators are estimated but omitted for parity with the main-text table layout. *Source:* Authors' own calculations based on CNMH/SIEVCAC microdata (Section S1). Software: Stata 19 (StataCorp, 2025).

S5.3.0.0 GAULA intervention and cumulative rescue

Unconditional margins from the Rescue Cox model show that GAULA intervention raises the predicted rescue probability at each duration category (Table S5.3).

Cumulative rescue probabilities by day 100 appear in Table S5.4 (exported in `outputs/Datos_Graficas_Cox.xlsx`).

Table S5.3: Rescue Probability by GAULA Intervention and Duration

GAULA intervention	Express	Short	Long
No	0.64	0.28	0.14
Yes	0.72	0.35	0.19

Note. Unconditional margins from the Rescue cause-specific Cox model estimated on the Cox subsample ($N = 461$). Duration categories: express (≤ 2 days), short (≤ 30 days), long (> 30 days). *Source:* Authors' own calculations based on CNMH/SIEVCAC microdata (Section S1). Software: Stata 19 (StataCorp, 2025).

Table S5.4: Cumulative Rescue Probability at Day 100

Perpetrator group	Pr(Rescue by day 100)
Paramilitaries	0.33
ELN	0.28
FARC	0.27

Note. Computed from cause-specific survival curves ($1 - S_t$) at day $t = 98$ (maximum grid point in `stcurve` with `range(0 100)`) estimated on the Cox subsample ($N = 461$) in `outputs/cumul_rescue_d100.dta`, exported to `outputs/Datos_Graficas_Cox.xlsx` by `microstructure_replication.do`. *Source:* Authors' own calculations based on CNMH/SIEVCAC microdata (Section S1). Software: Stata 19 (StataCorp, 2025).

S5.4.0.0 Mortality

The MNL yields unconditional average predicted death probabilities of 0.002 (FARC) and 0.071 (common delinquency) in Table S3.3. The corresponding cause-specific Cox margins reported in Section 4.1 of the main paper are 0.002 and 0.079 ($\approx$35$\times$ and $\approx$40$\times$, respectively). Both specifications preserve the same ordering: mortality rises as logistical retention capacity falls.

S6 Relevance for the mechanism

S6.0.0.1 Scope. This section maps the microeconometric results to the mechanism-design model (Section 4.1 of the main paper, "Empirical anchoring of the parameter space"). It introduces no new estimates. Every empirical quantity cited below is reported in Sections S1–S7 and can be verified from the replication package using `README.txt` at the package root.

Table S6.1: Mechanism Inputs and Replication Files

Mechanism object	Main-text result	Package file
Prior μ_0 over types θ_K	MNL outcome distribution	`microstructure_replication.do`
Terminal payment technology	RRR by victim profile	`sample_selection_robustness.do`
Arrival rates $\lambda_j(\theta_K, t)$	Cox cause-specific HRs	`microstructure_replication.do`
GAULA effectiveness γ_t	GAULA $\times$ duration margins	`outputs/Datos_Graficas_Cox.xlsx`
Sample validity	Attrition logit, IIA, robustness	`sample_selection_robustness.do`

Note. The MNL supplies the static type map; the Cox models supply dynamic resolution hazards that feed the daily public signal ζ_t in the MDG process. All paths are relative to `appendix_A_micro/scripts/`. *Source:* Authors' own calculations based on CNMH/SIEVCAC microdata (Section S1). Software: Stata 19 (StataCorp, 2025).

S7 Reproducibility

All numerical results in this supplement are reproduced by Stata 19 scripts in the `appendix_A_micro/scripts/` replication package. Run `run_all.do` from the `appendix\_A\_micro/scripts/` directory. See `README.md` for step-by-step instructions. Estimated runtime: 5–15 minutes. Batch mode: `stata -b do run_all.do`.

1. `run_all.do` runs `microstructure_replication.do`, `sample_selection_robustness.do`, and `cox_only.do` in sequence; all output is saved to `outputs/replication.log`.
2. Each script includes `_setup_paths.do` and locates `data/Data_merge.dta` automatically.
3. Key outputs written to `outputs/` include `analytical_sample.dta`; `cox_appendix_rows.txt` (Table S5.2); `cox_cumul_day100.txt` (Table S5.4); four `margins_*.dta` files (by zone, group, duration, profile); and `replication.log` (all model estimates and diagnostics).

4. `microstructure_replication.do` also exports `cumul_rescue_d100.dta` and `Datos_Graficas_Cox.xlsx` (Table S5.4); GAULA margins in Table S5.3 are recorded in `replication.log`.

5. `tables_mnl_full.tex` and `tables_cox_full.tex` (Tables S3.1 and S5.1) are preformatted LaTeX files; their entries are verified against `replication.log`.

6. (Optional) Compile the PDF: run `pdflatex` on `Appendix_A_Micro.tex` from the `appendix_A_micro/` directory.

Section S6 (Table S6.1) maps microeconometric outputs to mechanism inputs. Raw CNMH exports used to build `data/Data_merge.dta` are documented in `README.md`.

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